Lab 11: Proving the Backup

A definitive runbook for monthly operational validation.

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The Security Mandate

Reporter: Aisha (Security/GRC) | **Frequency:** Monthly Task

"Audit needs evidence our backups actually work. Do a real restore test."

The goal isn't just taking the snapshot—it's proving the data can be recovered.

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The Backup Typology Matrix

		
Automated Backups	**Manual Snapshots**	**AWS Backup**
Continuous, retention-bound system safety nets.	User-triggered, indefinite retention until explicitly deleted.	Centralized, policy-driven orchestration and compliance routing.

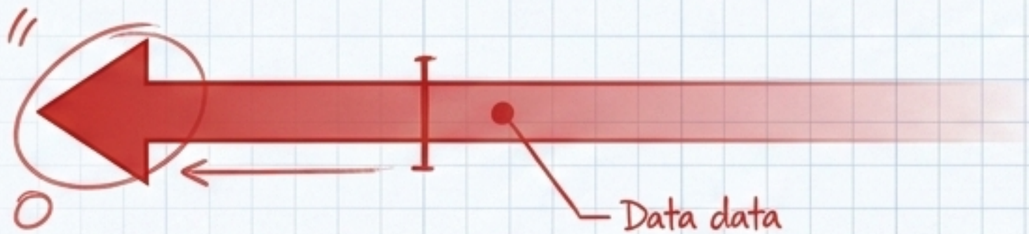
For this lab, we will generate a Manual Snapshot to isolate our test evidence.

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Defining Disaster Recovery Boundaries

RPO

Recovery Point Objective.

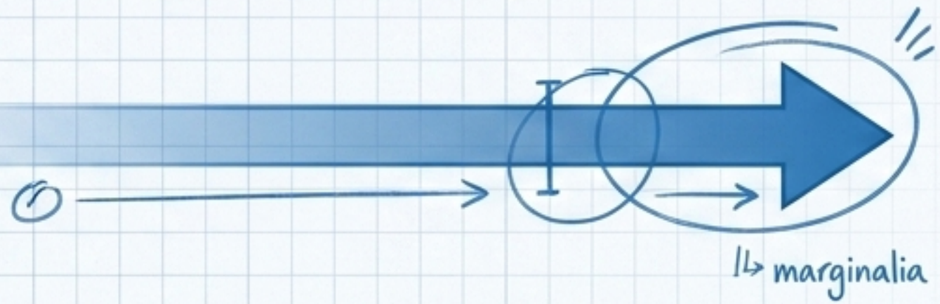


The data you can afford to lose. (Measured in historical time).

The Disaster Event

RTO

Recovery Time Objective.

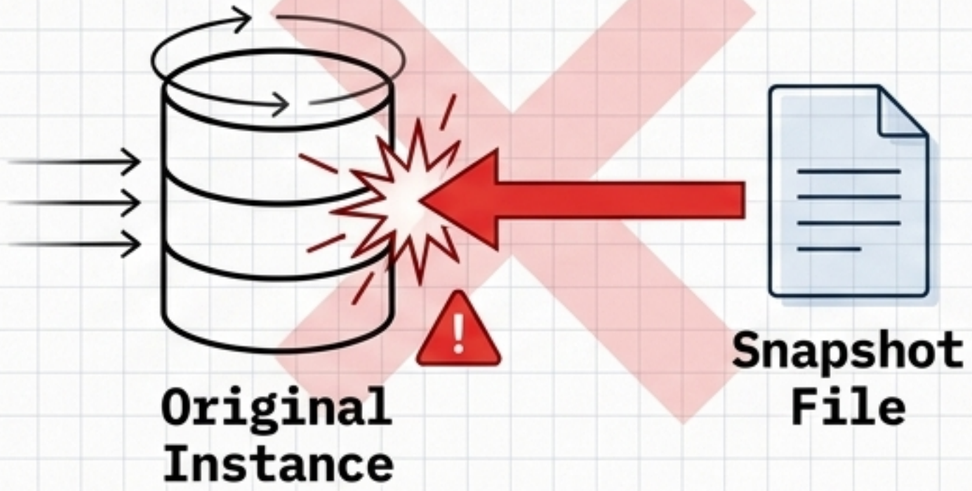


The time recovery may take. (Measured in future operational downtime).

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The Golden Rule of Restoration

The Wrong Way



The Right Way

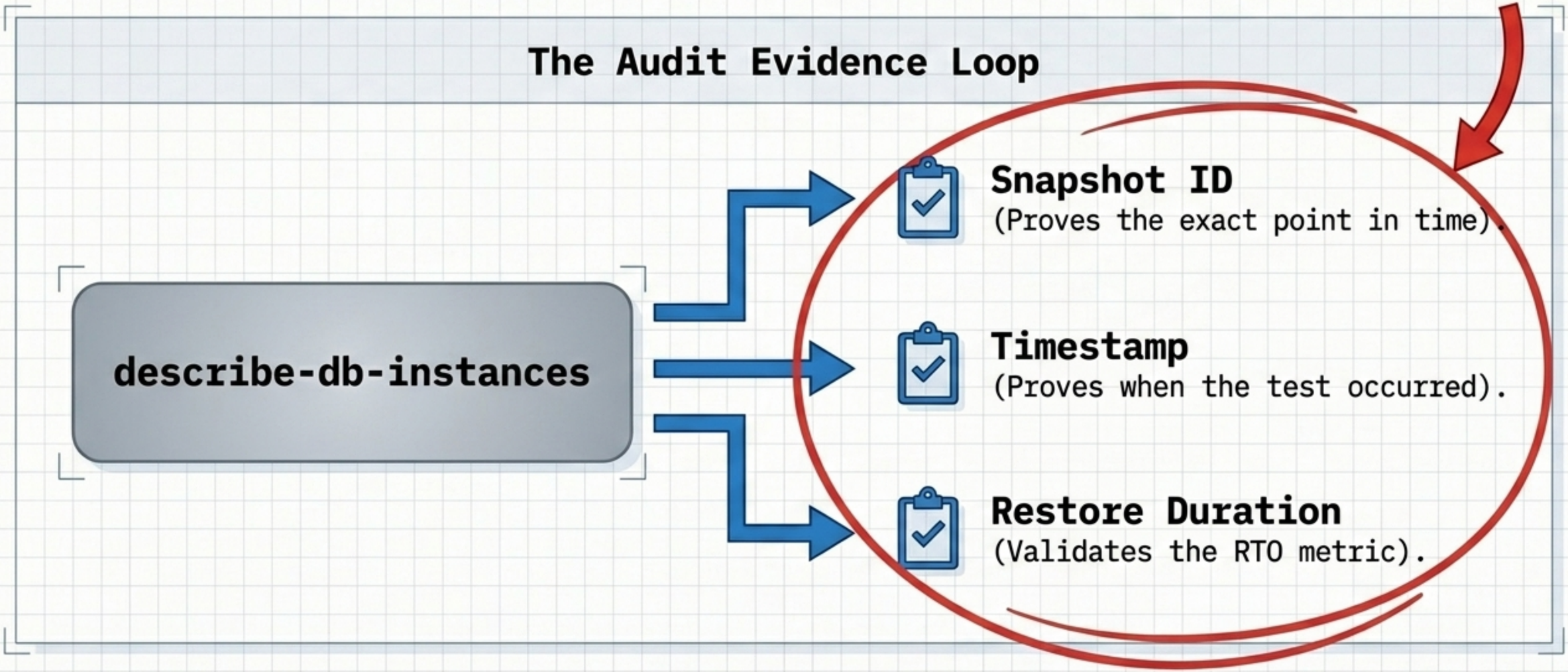


Always restore to a NEW instance — never over the original.

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The Audit Evidence Requirements

Capture this exact output to close Aisha's ticket.



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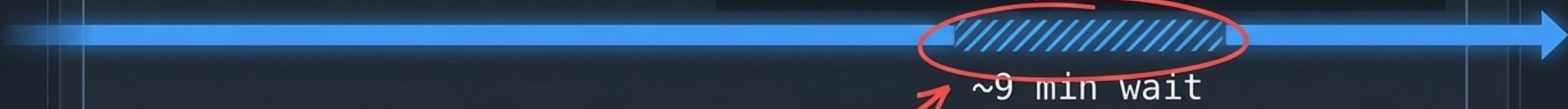
Execution Stage 1: State Capture

```
$ export DB_ID from terraform output
```

```
$ ./break.sh
```

Confirms backups are active and triggers the manual snapshot.

```
$ rds wait db-snapshot-available
```



Cut away. Don't watch it.
Let the ~9 min wait complete.

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Execution Stage 2: Parallel Restoration

```
$ restore-db-instance-from-db-snapshot

$ rds wait db-instance-available
```

Executing the Golden Rule: deploying to a new instance, never the original.

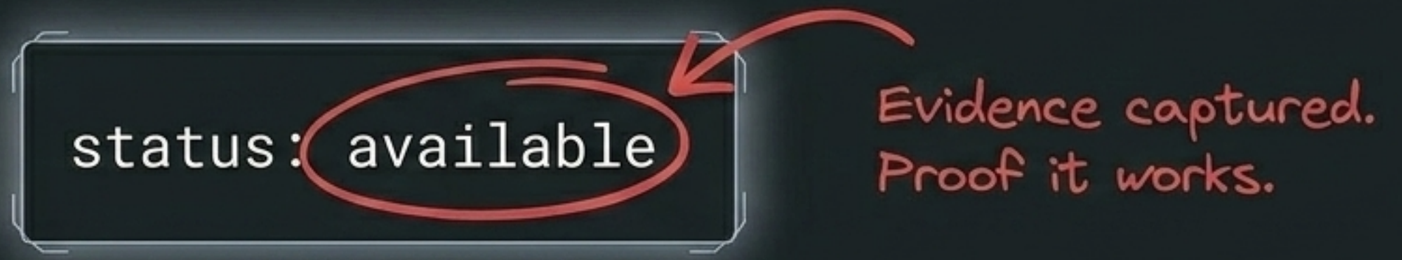


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Execution Stage 3: Verification

```
$ describe-db-instances
```

This command returns the Snapshot ID, Timestamp, and Restore Duration.



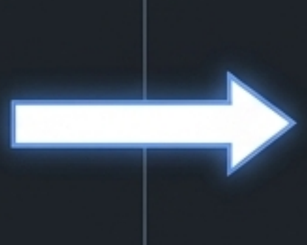
Success/Available

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The Cleanup Consequence

```
$ Delete the restored instance + the snapshot
```





$0

Mission Accomplished.
Zero cloud waste.





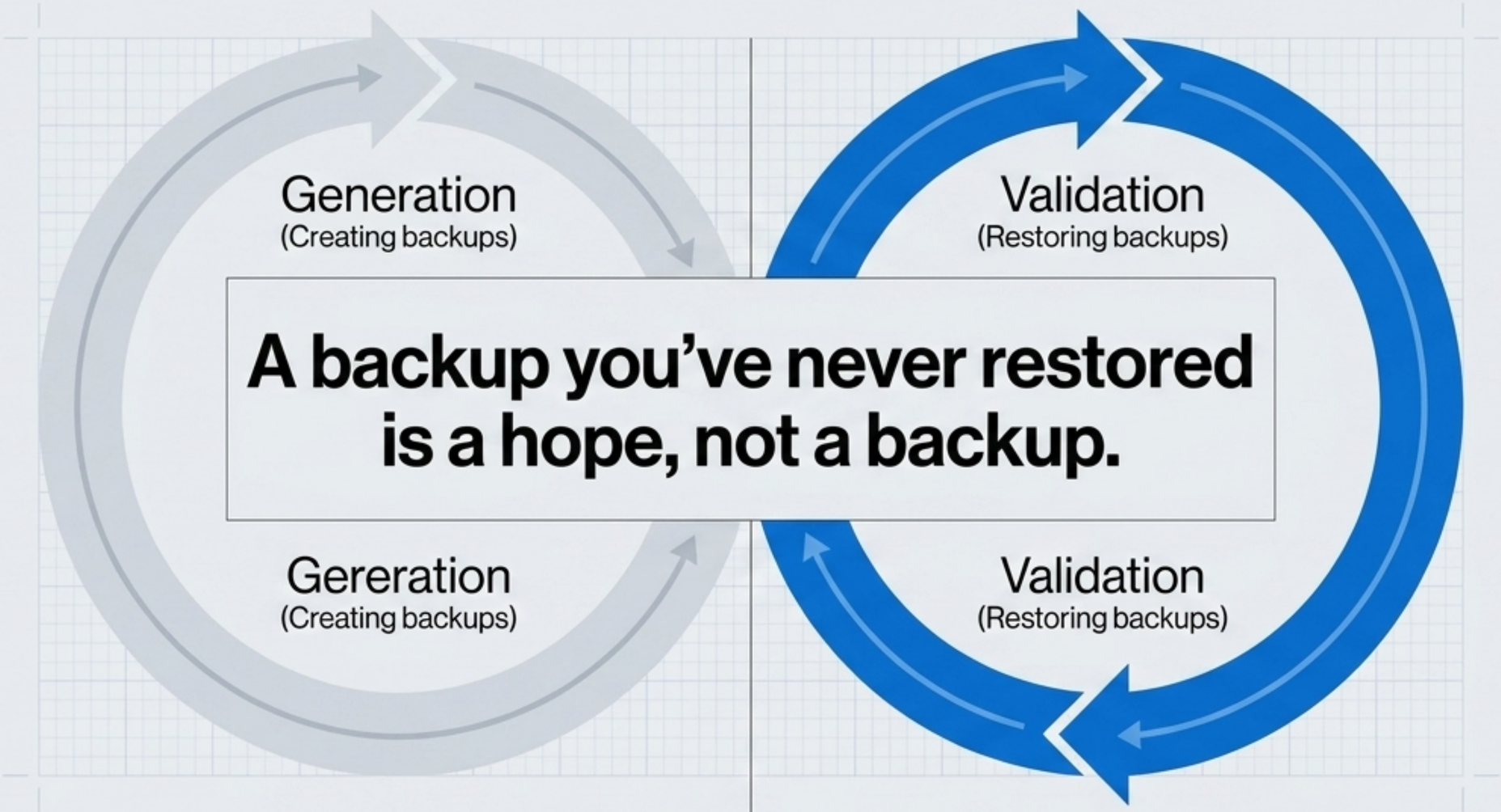


Real Cost

Orphaned resources result in compounding monthly cloud spend.

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The Day-2 Philosophy



This is a monthly loop. Schedule it. Never skip it.

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